

Recyc^{oo}l!

Wasted bits and bytes mean more heat, and an impediment to future progress in computing. Here's about "green computing"—where computers actually *recycle* bits!

Ram Mohan Rao

You can't teach your grandmother to suck eggs, a penny saved is a penny earned, and computers are getting faster every day. These are stock phrases, but they're facts of life. Computers are getting faster because of ever-increasing miniaturisation—packing in more transistors into a given amount of silicon real-estate, and other such approaches. This miniaturisation comes at a cost: heat. Chips are generating more heat, guzzling more electricity, and computers now account for almost 10 per cent of the power consumed in the United States—and 1.5 per cent globally, by some estimates.

At the current rate, we will not be able to keep up with the energy requirements of our computers; in addition, circuits simply *cannot* go small beyond a point, due to heat limitations. And as you know very well, the world now depends on ever-increasing computational power to sustain development.

The Point Of No Return

So just how far away is the point at which our computers won't be able to get any faster? It's way into the future. If you consider alternative computing paradigms, that is. Such as DNA computing (see *Digit*, November 2005), quantum computing, and others. But we're not talking about those things here. And we cannot talk about nanotech giving Moore's Law a new lease of life—because there is a fundamental physical limitation we must now talk about. Physics *guarantees* that a computer will not be able to discard

more than 3.5×10^{22} bits of physical information per second.

Umm...“discard”? Well, we're edging closer to our point. What's happening when your computer is doing its computations is, it's *throwing away* bits. For example, when you add 5 and 3 to get 8, the 5 and 3 are gone—forever. They're dissipated as heat. This is pretty simple to understand—after all, your processor gets hot because it's doing computations! Now think about it—can you possibly get back a 5 and a 3 from an 8? Given an 8, it could have been $1 + 7$, or $2 + 6$, or... in fact, it could have been any of an infinite number of combinations. So let's tie this up with what we said: our wasteful computers, working harder and harder, are throwing away more and more bits, but there's a limit. We won't name the limit because there are horrible equations involved—we'll just call it “the limit.”

To put the above figure into perspective, today's fastest processors are only about 1,00,000 times slower at discarding bits than the limit permits.

Isn't 1,00,000 large enough for, say, another 200 years? The simple answer is no. The researchers have it all worked out—they say that in 20 to 30 years, The Limit will have been reached. And that's in theory: in practice, The Limit will be reached even sooner. This is because of things such as thermal noise—the heat dissipated by the elements on an IC, because of the motion of electrons—which will become a concern in 20 years, and problems with leakage could come within ten years or less.

The answer? Obviously, stop throwing away the bits. Recycle them. “Uncompute”



Illustration pradip ingale